SECTION 26 29 11
MOTOR CONTROLLERS

PART 1 - GENERAL

1.1 DESCRIPTION

a. this section specifies the furnishing, installation, programming, integration, testing, and commissioning of low-voltage motor controllers and variable frequency drives (vfds) serving the hvac systems associated with this project, including:

1. air handling unit (ahu) supply fan motors.
2. primary and standby chilled water pumps.
3. any additional hvac equipment indicated on the contract documents as requiring variable speed control.

b. work includes combination motor controllers, magnetic starters, and low-voltage variable speed motor controllers (vfds), complete with disconnecting means, bypass assemblies, harmonic mitigation components, control transformers, and all accessories required for a complete and operational system.

c. all motor controllers and vfds furnished with mechanical equipment or provided as standalone assemblies shall comply with this section and shall be fully coordinated with division 22, division 23, and the existing building management system (bms).

d. vfds shall be fully integrated with the existing campus building management system and shall provide complete communication, monitoring, alarm reporting, and control capability in accordance with the sequence of operation.

e. this section does not apply to elevator motor controllers or fire pump controllers.

1.2 RELATED WORK

A. Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS: Requirements that apply to all sections of Division 26.

B. Section 26 05 13, MEDIUM-VOLTAGE CABLES: Medium-voltage cables and terminations.

C. Section 26 05 19, LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES: Low-voltage conductors.

D. Section 26 05 26, GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS: Requirements for personnel safety and to provide a low impedance path for possible ground fault currents.

E. Section 26 05 33, RACEWAY AND BOXES FOR ELECTRICAL SYSTEMS: Conduits.

F. Section 26 24 19, MOTOR CONTROL CENTERS: For multiple motor control assemblies which include motor controllers.

G. Section 02 82 11, TRADITIONAL ASBESTOS ABATEMENT.

1.3 QUALITY ASSURANCE

A. Quality Assurance shall be in accordance with Paragraph, QUALIFICATIONS (PRODUCTS AND SERVICES) in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS.

1.4 SUBMITTALS

A. Submit in accordance with Paragraph, SUBMITTALS in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS, and the following requirements:

1. 1. Shop Drawings:

   a. Submit sufficient information to demonstrate compliance with the Contract Documents for all motor controllers and variable frequency drives (VFDs) serving the AHU supply fan and chilled water pumps.

   b. Include the following for each VFD:

1. Manufacturer and model number.
2. Electrical ratings (HP, voltage, phase, FLA, short-circuit rating).
3. Enclosure type and environmental rating.
4. Integral input circuit breaker information.
5. Bypass configuration and wiring schematic.
6. Line reactor and harmonic mitigation components.
7. Dimensions, weights, and mounting details.
8. Heat rejection data.
9. Complete wiring diagrams and terminal identification.
10. Control interface details including 4–20 mA and/or 0–10 VDC input capability.
11. Network communication module details and protocol compatibility with existing Building Management System (BMS).

   c. Clearly identify compliance with IEEE 519 harmonic distortion limits.

   d. Clearly indicate coordination with the existing Building Management System and provide confirmation of communication compatibility.

2. Product Data:

   a. Submit manufacturer’s technical data sheets for VFDs, including performance curves, efficiency data, harmonic data, and overload characteristics.

   b. Submit documentation confirming suitability for variable torque applications (fans and pumps).

3. Coordination Submittal:

a. Submit written confirmation that the VFD manufacturer has coordinated with the existing controls contractor regarding communication protocol, integration requirements, and sequence of operation.

4. Certifications:
   a. Certification by the manufacturer that VFDs comply with Section 26 29 11 requirements.
   b. Certification that VFDs meet IEEE 519 harmonic limits at the point of common coupling.
5. Commissioning Documentation:
   a. Submit factory start-up report.
   b. Submit final programming parameters and commissioning checklist.
   c. Submit confirmation that final programming was performed by a factory-trained technician.

2. Manuals:
   a. Submit, simultaneously with the shop drawings, companion copies of complete maintenance and operating manuals, including technical data sheets, wiring diagrams, and information for ordering replacement parts.
      1) Wiring diagrams shall have their terminals identified to facilitate installation, maintenance, and operation.
      2) Wiring diagrams shall indicate internal wiring for each item of equipment and interconnections between the items of equipment.
      3) Elementary schematic diagrams shall be provided for clarity of operation.
      4) Include the catalog numbers for the correct sizes of overload relays for the motor controllers.
   b. If changes have been made to the maintenance and operating manuals originally submitted, submit updated maintenance and operating manuals two weeks prior to the final inspection.
3. Certifications: Two weeks prior to final inspection, submit the following.
   a. Certification by the manufacturer that the motor controllers conform to the requirements of the drawings and specifications.
   b. Certification by the Contractor that the motor controllers have been properly installed, adjusted, and tested.

1.5 APPLICABLE PUBLICATIONS

A. Publications listed below (including amendments, addenda, revisions, supplements, and errata) form a part of this specification to the extent referenced. Publications are referenced in the text by basic designation only.

B. Institute of Electrical and Electronic Engineers (IEEE):

519-14..................Recommended Practices and Requirements for Harmonic Control in Electrical Power Systems

C37.90.1-12.............Standard Surge Withstand Capability (SWC) Tests for Relays and Relay Systems Associated with Electric Power Apparatus

C. International Code Council (ICC):

IBC-21..................International Building Code

D. National Electrical Manufacturers Association (NEMA):

ICS 1-00(R2015).........Industrial Control and Systems: General Requirements

ICS 1.1-84(R2020).......Safety Guidelines for the Application, Installation and Maintenance of Solid State Control

ICS 2-00(R2020).........Industrial Control and Systems Controllers, Contactors, and Overload Relays Rated 600 Volts

ICS 4-15................Industrial Control and Systems: Terminal Blocks

ICS 6-93(R2016).........Industrial Control and Systems: Enclosures

ICS 7-20................Industrial Control and Systems: Adjustable-Speed Drives

ICS 7.1-14..............Safety Standards for Construction and Guide for Selection, Installation, and Operation of Adjustable-Speed Drive Systems

E. National Fire Protection Association (NFPA):

70-23...................National Electrical Code (NEC)

F. Underwriters Laboratories Inc. (UL):

508A-18.................Industrial Control Panels

1449-14.................Surge Protective Devices

61800-5-1-12............Adjustable Speed Electrical Power Drive Systems

PART 2 PRODUCTS

2.1 MOTOR CONTROLLERS

a. motor controllers shall comply with ieee, nema, nfpa 70, ul, and as shown on the drawings.

b. motor controllers shall be separately enclosed, unless part of another assembly. for installation in mechanical rooms, provide floor or wall mounted motor controllers as required by the installed environment.

c. motor controllers serving ahu-1 supply fan and chilled water pumps shall be combination type, with magnetic controller per paragraph 2.3 below and with circuit breaker disconnecting means, with external operating handle with lock-open padlocking positions and on-off position indicator.

d. enclosures:

1. enclosures shall be nema 1 for installation in conditioned mechanical spaces unless otherwise indicated on the drawings.

2. enclosure doors shall be interlocked to prevent opening unless the disconnecting means is open. provide padlocking provisions.

3. all metal surfaces shall be factory finished with manufacturer’s standard enamel finish.

e. motor control circuits shall operate at not more than 120 volts.

f. provide hand-off-automatic (h-o-a) switch for all non-vfd motor controllers unless specifically noted otherwise.

g. provide two normally open (n.o.) and two normally closed (n.c.) auxiliary contacts for each controller.

2.2 MANUAL MOTOR CONTROLLERS

manual motor controllers are not permitted for ahu-1 supply fan or chilled water pumps.

2.3 MAGNETIC MOTOR CONTROLLERS

magnetic motor controllers shall be provided only where indicated on the drawings and shall comply with paragraph 2.1 above.

2.4 REDUCED VOLTAGE MOTOR CONTROLLERS

reduced voltage motor controllers are not permitted unless specifically indicated on the drawings.

2.5 MEDIUM-VOLTAGE MOTOR CONTROLLERS

not applicable to this project.

2.6 LOW-VOLTAGE VARIABLE SPEED MOTOR CONTROLLERS (VSMC)

a. vsmc shall be provided for the following equipment:

1. ahu-1 supply fan motor.

2. p1 primary chilled water pump.

3. p2 standby chilled water pump.

b. vsmc shall be abb acs580-01 series, manufactured in the united states, ul listed, and compliant with nema ics 7.

c. vsmc shall be electronic, with adjustable frequency and voltage, three phase output, capable of driving standard nema b three-phase induction motors at full rated speed. the control technique shall be pulse width modulation (pwm). silicon controlled rectifiers or other control techniques are not acceptable.

d. vsmc shall be suitable for variable torque loads associated with hvac fans and centrifugal pumps.

e. vsmc shall be rated for the motor horsepower and full load amperes as indicated on the drawings.

f. vsmc current and voltage harmonic distortion shall not exceed the values allowed by ieee 519. provide integral 5 percent line reactor.

g. vsmc shall include an input circuit breaker interlocked with the enclosure door.

h. provide contactor-style bypass for ahu-1 supply fan and primary chilled water pump vfds. bypass shall include motor overload protection and h-o-a control.

i. vsmc shall accept 4-20 ma or 0-10 vdc control signals from the existing building management system.

j. vsmc shall provide the following feedback points to the existing building management system:

1. run status.

2. fault status.

3. output frequency.

4. motor current.

k. provide all hardware, software, network interfaces, gateways, and programming required to integrate vsmc with the existing va building management system.

l. final programming and commissioning shall be performed by a factory-trained technician in coordination with the existing controls contractor.

PART 3 EXECUTION

3.1 INSTALLATION

a. install motor controllers and vsmc in accordance with nec and manufacturer’s recommendations.

b. install vsmc adjacent to equipment served unless otherwise approved.

c. program vsmc per manufacturer’s instructions and in coordination with the existing building management system.

3.2 ACCEPTANCE CHECKS AND TESTS

a. perform manufacturer’s required field tests.

b. for vsmc, final programming and startup shall be performed by factory-trained technician.

3.3 FOLLOW-UP VERIFICATION

a. demonstrate proper operation of ahu-1 and chilled water pump vfds under all operating modes, including bypass operation.

3.4 SPARE PARTS

provide one spare set of fuses for each combination motor controller.

3.5 INSTRUCTION

provide two 4-hour training sessions for va personnel covering operation and maintenance of vsmc.

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